

AI



SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: B Tech/MBA Tech

Year: II Semester: IV

Stream/s : AI/AIDS/AIML/CSE(DS) _____

Subject: Machine Learning

Time: 3 hrs (10.00 am to 1.00 pm)

Date: 18/04/2024

No. of Pages: 5

Marks: 100

Final Examination/Re-Examination (2022-2023)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly (Each question carries 4 marks):	[20]										
CO-1; SO-1; BL-1	a.	Predicting the price of a pre-owned vehicle depends on the number of factors such as Purchased year, Current showroom price, Fuel type, kilometers driven, Existing owners, Transmission type, Seller type. Answer the following for this application: 1. Identify the appropriate category (supervised/unsupervised) of the machine learning algorithm that can be applied to above task 2. Identify a suitable algorithm which can be used for this task 3. List dependent and independent features.											
CO-1; SO-2; BL-3	b.	Identify outliers from the following quiz scores for 18 students using boxplot: (11, 8, 12, 12, 5, 13, 12, 14, 14, 13, 13, 14, 13, 15, 15, 15, 15)											
CO-2; SO-2; BL-5	c.	<table><tr><th>x (No. of years in the organization)</th><th>y (skill upgradation courses completed)</th></tr><tr><td>3</td><td>1</td></tr><tr><td>4</td><td>2</td></tr><tr><td>5</td><td>2</td></tr><tr><td>6</td><td>3</td></tr></table> Evaluate the updated weights (parameters) after one iteration if simple linear regression is applied by using gradient descent. Use learning rate $\alpha = 0.05$ and initial weights, $\theta_0 = \theta_1 = 0.5$.	x (No. of years in the organization)	y (skill upgradation courses completed)	3	1	4	2	5	2	6	3	
x (No. of years in the organization)	y (skill upgradation courses completed)												
3	1												
4	2												
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6	3												
CO-2; SO-2; BL-2	d.	Explain the significance of sigmoid function in logistic regression. Justify its use for logistic regression.											

CO-3; SO-2; BL-3	e.	A classifier is used to train a machine to classify cat, dog and Monkey. Confusion matrix for this classifier is shown below. Determine recall for each class.																																													
		<table><tr><td></td><td></td><th colspan="3">Predicted</th></tr><tr><td></td><td></td><th>Cat</th><th>Dog</th><th>Monkey</th></tr><tr><th rowspan="3">Actual</th><th>Cat</th><td>18</td><td>6</td><td>5</td></tr><tr><th>Dog</th><td>4</td><td>20</td><td>4</td></tr><tr><th>Monkey</th><td>3</td><td>4</td><td>30</td></tr></table>			Predicted					Cat	Dog	Monkey	Actual	Cat	18	6	5	Dog	4	20	4	Monkey	3	4	30																						
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Q2																																															
CO-1; SO-2; BL-3	A	The dataset given below is for the land price detection model for various zones in a city using Area of the land and Air Quality Index. Apply normalization scaling technique on the following two features: 1. Area of the Land 2. Air Quality Index <table><tr><th>Sr No</th><th>Area of the Land (sq. meters)</th><th>Air Quality Index (AQI)</th><th>Land Price (Rs. Per acre)</th></tr><tr><td>1.</td><td>13690</td><td>146</td><td>120000</td></tr><tr><td>2.</td><td>12555</td><td>42</td><td>150000</td></tr><tr><td>3.</td><td>11900</td><td>139</td><td>345000</td></tr><tr><td>4.</td><td>12560</td><td>144</td><td>445000</td></tr><tr><td>5.</td><td>23000</td><td>37</td><td>600000</td></tr><tr><td>6.</td><td>35550</td><td>350</td><td>350000</td></tr><tr><td>7.</td><td>42000</td><td>45</td><td>450000</td></tr><tr><td>8.</td><td>45000</td><td>112</td><td>550000</td></tr><tr><td>9.</td><td>27700</td><td>77</td><td>250000</td></tr><tr><td>10.</td><td>12345</td><td>82</td><td>377133</td></tr></table>	Sr No	Area of the Land (sq. meters)	Air Quality Index (AQI)	Land Price (Rs. Per acre)	1.	13690	146	120000	2.	12555	42	150000	3.	11900	139	345000	4.	12560	144	445000	5.	23000	37	600000	6.	35550	350	350000	7.	42000	45	450000	8.	45000	112	550000	9.	27700	77	250000	10.	12345	82	377133	[5]
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CO-1; SO-2; BL-2	B	Explain the term ‘curse of dimensionality’. What problem does it cause? List down the methods used in Dimensionality Reduction.	[5]																																												
CO-3; SO-2; BL-4	C	1. Use the K nearest neighbor classifier (KNN) to categorize the patient with BMI=43.6 and Age=40 as diabetic/non-diabetic. Given the training examples: (Assume K=3 and use Euclidean Distance) 2. What will be the classification if k=5? How do we choose the optimum value of K in KNN algorithm? <table><tr><th>BMI</th><th>Age</th><th>Diabetic/non-diabetic</th></tr><tr><td>33.6</td><td>50</td><td>1</td></tr><tr><td>26.6</td><td>30</td><td>0</td></tr><tr><td>23.4</td><td>40</td><td>0</td></tr><tr><td>43.1</td><td>67</td><td>0</td></tr><tr><td>35.3</td><td>23</td><td>1</td></tr><tr><td>35.9</td><td>67</td><td>1</td></tr></table>	BMI	Age	Diabetic/non-diabetic	33.6	50	1	26.6	30	0	23.4	40	0	43.1	67	0	35.3	23	1	35.9	67	1	[10]																							
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Q3																																															

Q3

CO-2; SO-2; BL-2	A	1. With the help of an example, develop the model for multiple linear regression: Outline the steps for estimating the parameters of the model. 2. Justify the need of polynomial regression giving an example.	[10]																																																	
CO-3; SO-2; BL-4	B	Consider the dissimilarity matrix for the data points. Perform agglomerative clustering on the data points and draw the dendrogram. Use Euclidean distance as a distance measure. <table border="1"><tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td></tr><tr><td>A</td><td>0</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>B</td><td>2</td><td>0</td><td></td><td></td><td></td><td></td></tr><tr><td>C</td><td>4</td><td>3</td><td>0</td><td></td><td></td><td></td></tr><tr><td>D</td><td>10</td><td>7</td><td>9</td><td>0</td><td></td><td></td></tr><tr><td>E</td><td>8</td><td>5</td><td>6</td><td>3</td><td>0</td><td></td></tr><tr><td>F</td><td>4</td><td>6</td><td>5</td><td>2</td><td>1</td><td>0</td></tr></table>		A	B	C	D	E	F	A	0						B	2	0					C	4	3	0				D	10	7	9	0			E	8	5	6	3	0		F	4	6	5	2	1	0	[10]
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CO-1; SO-2; BL-2	A	Explain and illustrate with examples the various techniques that can be used to handle missing data in machine learning.	[5]																																																	
CO-3; SO-2; BL-3	B	1. Estimate the equation of the line separating positively labelled data points (1,1), (2,0), (1, -1) and negatively labelled data points (4,0), (5,1), (5, -1) using Support Vector Machine (SVM) algorithm. 2. How is SVM used for classifying non-linearly separable data? Explain what is the kernel trick in SVM.	[10]																																																	
CO-3; SO-2; BL-2	C	What is ensemble learning, and how does it differ from traditional machine learning approaches?	[5]																																																	
Q5																																																				
CO-2; SO-2; BL-3	A	The dataset given below shows the relationship between the number of hours spent on writing essay and the grade scored. Determine the best fitting line using ordinary least square regression (OLS). <table border="1"><tr><th>Sr No</th><th>Hours spent on Essay</th><th>Grade</th></tr><tr><td>1</td><td>6</td><td>82</td></tr><tr><td>2</td><td>10</td><td>88</td></tr><tr><td>3</td><td>2</td><td>56</td></tr><tr><td>4</td><td>4</td><td>64</td></tr><tr><td>5</td><td>6</td><td>77</td></tr><tr><td>6</td><td>7</td><td>92</td></tr><tr><td>7</td><td>0</td><td>23</td></tr><tr><td>8</td><td>1</td><td>41</td></tr><tr><td>9</td><td>8</td><td>80</td></tr><tr><td>10</td><td>5</td><td>59</td></tr><tr><td>11</td><td>3</td><td>47</td></tr></table>	Sr No	Hours spent on Essay	Grade	1	6	82	2	10	88	3	2	56	4	4	64	5	6	77	6	7	92	7	0	23	8	1	41	9	8	80	10	5	59	11	3	47	[10]													
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CO-3; SO-2; BL-3	B	Consider the data for the six cites shown in the table, where number of schools and the student strength in each school is shown. Use K means algorithm to cluster the data and show the updated cluster centers after first iteration. Assume k = 2 and (2,2) and (3,3) as initial cluster centers.	[10]																																													
		<table><tr><th>Sr No</th><th>Number of schools</th><th>student strength in each school (in thousands)</th></tr><tr><td>1</td><td>3</td><td>3</td></tr><tr><td>2</td><td>2</td><td>2</td></tr><tr><td>3</td><td>2</td><td>4</td></tr><tr><td>4</td><td>2</td><td>1</td></tr><tr><td>5</td><td>4</td><td>2</td></tr><tr><td>6</td><td>3</td><td>1</td></tr></table>		Sr No	Number of schools	student strength in each school (in thousands)	1	3	3	2	2	2	3	2	4	4	2	1	5	4	2	6	3	1																								
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CO-3; SO-2; BL-4	A	A company is trying to predict whether customers will purchase a product based on demographic and purchase history data. They have collected the following dataset:	[10]																																													
		<table><tr><th>Customer</th><th>Age</th><th>Gender</th><th>Income</th><th>Purchase</th></tr><tr><td>1</td><td>Young</td><td>Male</td><td>Low</td><td>No</td></tr><tr><td>2</td><td>Young</td><td>Male</td><td>High</td><td>No</td></tr><tr><td>3</td><td>Middle-aged</td><td>Male</td><td>Low</td><td>Yes</td></tr><tr><td>4</td><td>Senior</td><td>Male</td><td>High</td><td>Yes</td></tr><tr><td>5</td><td>Senior</td><td>Female</td><td>Low</td><td>Yes</td></tr><tr><td>6</td><td>Middle-aged</td><td>Female</td><td>Low</td><td>Yes</td></tr><tr><td>7</td><td>Young</td><td>Female</td><td>High</td><td>No</td></tr><tr><td>8</td><td>Young</td><td>Female</td><td>Low</td><td>Yes</td></tr></table>		Customer	Age	Gender	Income	Purchase	1	Young	Male	Low	No	2	Young	Male	High	No	3	Middle-aged	Male	Low	Yes	4	Senior	Male	High	Yes	5	Senior	Female	Low	Yes	6	Middle-aged	Female	Low	Yes	7	Young	Female	High	No	8	Young	Female	Low	Yes
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Using entropy as the splitting criterion, decide which feature (Age/ Gender/ Income) should be used as the root node for splitting the samples.																																																

CO-3; SO-2; BL-3	B	1. What is the Naïve assumption in Naïve Bayes classifier? 2. Given a dataset to predict the species from the features, Predict the species of an entity with the following attributes using the Naïve Bayes classification approach, $X = \{\text{Legs}=2, \text{Height}=\text{Tall}, \text{Smelly}=\text{No}\}$	[10]																																	
		<table border="1"> <thead> <tr> <th>Legs</th><th>Height</th><th>Smelly</th><th>Species</th></tr> </thead> <tbody> <tr><td>3</td><td>Short</td><td>Yes</td><td>M</td></tr> <tr><td>2</td><td>Tall</td><td>No</td><td>M</td></tr> <tr><td>3</td><td>Short</td><td>Yes</td><td>M</td></tr> <tr><td>3</td><td>Short</td><td>Yes</td><td>M</td></tr> <tr><td>2</td><td>Short</td><td>No</td><td>H</td></tr> <tr><td>2</td><td>Tall</td><td>No</td><td>H</td></tr> <tr><td>2</td><td>Tall</td><td>No</td><td>H</td></tr> <tr><td>2</td><td>Short</td><td>Yes</td><td>H</td></tr> </tbody> </table>		Legs	Height	Smelly	Species	3	Short	Yes	M	2	Tall	No	M	3	Short	Yes	M	3	Short	Yes	M	2	Short	No	H	2	Tall	No	H	2	Tall	No	H	2
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Q7																																				
CO-1; SO-1; BL-2	A	1. Compare Parametric and non-parametric machine learning models. 2. Explain briefly the points which influence the choice of Parametric and non-parametric machine learning models?	[4]																																	
CO-3; SO-2; BL-2	B	1. Compare bagging and boosting techniques used in ensemble learning. 2. Explain the basic concept of Adaboost classifier.	[8]																																	
CO-3; SO-2; BL-2	C	With respect to DBSCAN algorithm, explain the following terms with suitable examples: 1. Core points 2. Border points 3. Density reachable points 4. Outliers	[8]																																	